Lalit Pathak, Ph.D.

Gravitational-Wave Astronomy · GW Data Analysis · Compact Binaries · Fundamental Physics

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Research Summary

My research focuses on extracting astrophysical and fundamental-physics information from gravitational-wave observations. I develop and apply computational and statistical methods for gravitational-wave data analysis, with particular interests in compact-binary parameter estimation, waveform modelling, population inference, gravitational-wave cosmology, tests of general relativity, and the development of methods for current and future detector networks.

Research Experience

Postdoctoral Fellow Aug 2025 – Present

IUCAA, Pune, India

- Investigating the impact of gravitational-wave lensing on compact-binary parameter inference, including degeneracies between lensing effects and intrinsic source parameters.
- Developing statistical methods to extract information about the population and redshift distribution of binary-black-hole mergers from gravitational-wave observations.
- Developing machine-learning approaches for estimating the stochastic gravitational-wave background and studying intermediate-mass binary-black-hole mergers in the decihertz band.
- Leading the PyCBC-based subsolar-mass compact-binary search for the third part of the fourth LIGO–Virgo–KAGRA observing run, and contributing to low-latency gravitational-wave searches using PyCBC Live on IUCAA's dedicated search infrastructure.
- Contributing to and reviewing LIGO–Virgo–KAGRA scientific analyses.

Postdoctoral Fellow Jul 2024 – Jul 2025

Department of Astronomy and Astrophysics, TIFR, Mumbai, India

- Assessed systematic effects of compact-binary delay-time distribution models on the astrophysical and cosmological properties inferred from gravitational-wave populations.
- Contributed to LIGO–Virgo–KAGRA analyses of compact-binary mass distributions and their astrophysical interpretation, and to the development and review of population-inference and gravitational-wave astrophysics analyses.
- Contributed to mock-injection and parameter-estimation analyses for a blinded spectral-siren measurement of the Hubble constant within the LIGO–Virgo–KAGRA gravitational-wave cosmology program.

Visiting Research Assistant Feb 2024 – Jul 2024

Department of Astronomy and Astrophysics, TIFR, Mumbai, India

- Initiated a study of how compact-binary delay-time distribution models affect astrophysical and cosmological inference from gravitational-wave populations.
- Contributed to LIGO–Virgo–KAGRA scientific studies involving compact-binary populations and gravitational-wave inference.

Ph.D., Physics 2018 – 2024

Indian Institute of Technology Gandhinagar, Gandhinagar, India

- Developed a fast parameter-estimation framework for compact-binary coalescences based on numerical linear algebra and meshfree approximations, enabling rapid evaluation of source-parameter posteriors from gravitational-wave data.
- Extended the framework from single-detector inference to coherent multi-detector parameter estimation, incorporating sky localization and other extrinsic parameters.
- Applied the framework to large-scale simulations to investigate the impact of LIGO-India on the characterization and localization of compact-binary sources.
- Developed a fast and faithful meshfree interpolation of numerical-relativity surrogate waveforms for efficient waveform evaluation in large-scale parameter-estimation studies.
- Collaborated on tests of the second postulate of relativity using the GW170817 binary-neutron-star event, constraining phenomenological deviations from the standard relativistic propagation model.
- Contributed to LIGO–Virgo–KAGRA scientific studies during the fourth observing run.

M.Sc. Dissertation Jan 2016 – Jun 2016

Department of Physics and Astrophysics, University of Delhi, Delhi, India

- Studied Bose–Einstein condensation and its possible role in the formation of supermassive black holes, including effects of large extra dimensions on Bose–Einstein condensates.

Summer Research Fellow May 2013 – Jul 2013

Physical Research Laboratory (PRL), Ahmedabad, India

- Completed a summer research project on the Standard Model of Particle Physics under the mentorship of Prof. Utpal Sarkar.

Education

Ph.D. in Physics — Indian Institute of Technology Gandhinagar 2024

M.Sc. in Physics — Department of Physics and Astrophysics, University of Delhi 2017

B.Sc. (Hons.) in Physics — Hansraj College, University of Delhi 2013

Selected Research Publications

1. **L. Pathak**, H. Phurailatpam, and A. Gopakumar, "On the Presence of a Tertiary Compact Object in GW190814," arXiv:2605.21955 (2026), accepted in *ApJ*.
2. A. Sharma, **L. Pathak**, S. Roy, and A. S. Sengupta, "Rapid parameter estimation with the full symphony of compact binary mergers using meshfree approximation," *Phys. Rev. D* **114**, 044043 (2026), arXiv:2508.04172 (2025).
3. C. Karathanasis, S. Mukherjee, **L. Pathak**, S. Vallejo-Peña, M. R. Sah, B. Revenu, A. E. Romano, and J. Garcia-Bellido, "Blinded Mock Data Challenge: Is the Spectral Siren Technique Robust for Measuring the Hubble Constant?," arXiv:2604.26273 (2026).
4. **L. Pathak**, A. Reza, and A. S. Sengupta, "Fast and faithful interpolation of numerical relativity surrogate waveforms using meshfree approximation," *Phys. Rev. D* **110**, 064022 (2024).
5. S. Shukla, **L. Pathak**, and A. Sengupta, "Pinpointing Binary Neutron Star sources with a global network of GW Detectors, including LIGO-Aundha," *Phys. Rev. D* **109**, 044051 (2024).
6. **L. Pathak**, S. Munishwar, A. Reza, and A. S. Sengupta, "Prompt sky localization of compact binary coalescences using meshfree approximation," *Phys. Rev. D* **109**, 024053 (2024).
7. R. Ghosh, S. Nair, **L. Pathak**, S. Sarkar, and A. S. Sengupta, "Does the speed of gravitational waves depend on the source velocity?," *Phys. Rev. D* **108**, 124017 (2023).
8. **L. Pathak**, A. Reza, and A. S. Sengupta, "Fast likelihood evaluation using meshfree approximations for reconstructing compact binary sources," *Phys. Rev. D* **108**, 064055 (2023).

LIGO–Virgo–KAGRA Collaboration Publications

1. LIGO Scientific and KAGRA and Virgo Collaborations, A. G. Abac et al., "Updated Upper Limits on the Isotropic Gravitational-Wave Background from LIGO, Virgo, and KAGRA Data through April 2025," arXiv:2608.23477 (2026).
2. LIGO Scientific and KAGRA and Virgo Collaborations, A. G. Abac et al., "Constraints on ultralight bosons from merging binary and remnant black holes observed during the second and third parts of the fourth LIGO-Virgo-KAGRA observing run," arXiv:2608.11620 (2026).
3. L. Sun et al., LIGO Scientific and KAGRA and Virgo Collaborations, "LIGO A♯: Detector Design and Science Prospects Beyond A+," arXiv:2608.11673 (2026).
4. LIGO Scientific and KAGRA and Virgo Collaborations, A. G. Abac et al., "GWTC-5.0: Tests of General Relativity," arXiv:2607.19293 (2026).
5. LIGO Scientific and KAGRA and Virgo Collaborations, A. G. Abac et al., "Open Data from LIGO, Virgo, and KAGRA through the Second Part of the Fourth Observing Run," arXiv:2605.27090 (2026).
6. LIGO Scientific and KAGRA and Virgo Collaborations, A. G. Abac et al., "GWTC-5.0: Observations from the Second Part of the Fourth LIGO-Virgo-KAGRA Observing Run and Updates to the Gravitational-Wave Transient Catalog," arXiv:2605.27225 (2026).
7. LIGO Scientific and KAGRA and Virgo Collaborations, A. G. Abac et al., "GWTC-5.0: Population Properties of Merging Compact Binaries," arXiv:2605.27226 (2026).
8. LIGO Scientific and KAGRA and Virgo Collaborations, A. G. Abac et al., "GWTC-5.0: Constraints on the Cosmic Expansion Rate and Modified Gravitational-wave Propagation," arXiv:2605.27227 (2026).
9. LIGO Scientific and KAGRA and Virgo Collaborations, A. G. Abac et al., "GWTC-5.0: Methods for Identifying and Characterizing Gravitational-wave Transients," arXiv:2605.27224 (2026).
10. LIGO Scientific and KAGRA and Virgo Collaborations, A. G. Abac et al., "GWTC-5.0: An Introduction to Version 5.0 of the Gravitational-Wave Transient Catalog," arXiv:2605.27223 (2026).
11. LIGO Scientific and KAGRA and Virgo Collaborations, A. G. Abac et al., "GWTC-4.0: Updating the Gravitational-Wave Transient Catalog with Observations from the First Part of the Fourth LIGO-Virgo-KAGRA Observing Run," arXiv:2508.18082 (2025).
12. LIGO Scientific and KAGRA and Virgo Collaborations, A. G. Abac et al., "GWTC-4.0: Population Properties of Merging Compact Binaries," arXiv:2508.18083 (2025).
13. LIGO Scientific and KAGRA and Virgo Collaborations, A. G. Abac et al., "GWTC-4.0: Methods for Identifying and Characterizing Gravitational-wave Transients," arXiv:2508.18081 (2025).
14. LIGO Scientific and KAGRA and Virgo Collaborations, A. G. Abac et al., "GWTC-4.0: An Introduction to Version 4.0 of the Gravitational-Wave Transient Catalog," *Astrophys. J. Lett.* **995** (2025) 1, L18.

Selected Talks & Conference Presentations

- "Rapid construction of compact binary sources using meshfree approximation," AAPCOS 2023, oral presentation, SINP, Kolkata.
- "Rapid construction of compact binary sources using meshfree approximation," GWPAW 2022, poster presentation, Melbourne, Australia.
- "Meshfree interpolation of likelihood for fast parameter estimation of CBCs," AAPS-DACG Workshop, 2021.
- "Meshfree interpolation of likelihood for fast parameter estimation of CBCs," GWPAW 2021, poster presentation (online), Hannover, Germany.

Selected Schools & Workshops

- GW Summer School 2023, ICTS, Bengaluru — Numerical Relativity.
- GW Summer School 2022, ICTS, Bengaluru — Astrophysics of GW sources, structure and evolution of stars, and evolutionary history of compact binaries.
- Newton–Bhabha Workshop on Gravitational Waves Astronomy, IUCAA, Pune (2019).

Technical Skills

Programming	Python, C, C++
Gravitational-wave analysis	Parameter estimation, likelihood evaluation, coherent network inference, sky localization, waveform modelling
Computational methods	Numerical linear algebra, meshfree approximation, interpolation, large-scale simulation
Research applications	Compact-binary coalescences, stochastic gravitational-wave background, population inference, gravitational-wave cosmology, Machine learning

Awards

- Tata Consultancy Services (TCS) Research Scholarship, 2019 — awarded to support Ph.D. research at IIT Gandhinagar.

References

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Academy Scientist in Machine Learning, Österreichische Akademie der Wissenschaften, Austria
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Professor, Indian Institute of Technology Gandhinagar, India
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